MVP User Stories & Acceptance Criteria

JIRA: MEAL-3 | Version: 1.0 | Project: Meal Connect | Team: Apex

Implementation-ready story catalog for donors (restaurants/bakeries), verified NGOs, collectors/volunteers and administrators. Linked to requirement identifiers for testing and delivery planning.

1. Purpose and boundaries

This guide defines observable user behaviors, objective pass/fail acceptance conditions and traceability to the Software Requirements Specification (ACC, LIS, REQ, FUL, SEC, USA, REL, TRA). It reflects the defined MVP, not a claim that these features have already been implemented or tested.

Technology baseline: Angular 21 + TypeScript + SCSS frontend; Python FastAPI REST backend; PostgreSQL; local development without Docker. Detailed API contracts: MEAL-62 v1.3. Data model: MEAL-6. Architecture: MEAL-5 and MEAL-61 v1.3.

2. Non-negotiable MVP rules

- Only verified NGOs can request quantities from existing, available donation listings. NGOs do not publish independent food-need posts.
- A business can approve the whole requested quantity or reject the request; no partial approval and no approved_quantity field.
- Approval must atomically reserve the requested quantity without exceeding the donation total, including under concurrent requests.
- Only authorized users with suitable active organization membership can modify organization-scoped records.
- Pickups and distribution records follow approved requests; the final distribution record is traceable to a pickup and donation.
- Food safety cannot be guaranteed by software. Sri Lankan food-handling requirements need qualified verification before a real-world pilot.

3. User story register

4. User stories and objective acceptance conditions

US-01 | Register account

Role: All users Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a prospective user, I want to register so that I can access appropriate platform services.

Requirement links: ACC-001, ACC-003, ACC-005; SEC-001; REL-001

- AC-01: Given valid required registration data and a previously unused email, when registration is submitted, then a user record is created and a success response is returned.
- AC-02: Given an email already associated with an account, when registration is submitted, then the system rejects the duplicate without creating a second account.

- AC-03: The stored credential is a secure password hash; the plaintext password is not persisted.

US-02 | Sign in and protect sessions

Role: All users Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a registered user, I want to sign in securely so that I can access my authorized workspace.

Requirement links: ACC-002, ACC-006; SEC-001, SEC-002, SEC-003

- AC-01: Given valid active-account credentials, when the login request succeeds, then a bearer access token is returned according to MEAL-62.
- AC-02: Given invalid credentials or a disabled account, when login is attempted, then access is denied without exposing credentials.
- AC-03: Given a missing/invalid token on a protected operation, when the API is called, then the backend returns an unauthorized response.

US-03 | Submit organization profile

Role: Restaurant / NGO Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an authorized business or NGO user, I want to register an organization profile so that it can be verified.

Requirement links: ACC-004, ACC-005; SEC-003

- AC-01: When a permitted user submits the required organization information, the system stores a profile with an initial verification state.
- AC-02: A user without permission cannot edit another organization's details.
- AC-03: An organization that is not approved cannot perform operations reserved for verified businesses or NGOs.

US-04 | Verify organization

Role: Administrator Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an administrator, I want to approve or reject organization verification so that only eligible partners operate.

Requirement links: ACC-008; SEC-002, SEC-003; TRA-003

- AC-01: Given a pending organization, an authorized administrator can record an approval or rejection decision.
- AC-02: Each verification decision records the reviewing actor and decision time.
- AC-03: Non-administrators cannot perform verification decisions through the API.

US-05 | Create donation listing

Role: Restaurant / Bakery Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As business staff, I want to post surplus food so that NGOs can find it before the pickup deadline.

Requirement links: LIS-001, LIS-002, LIS-003, LIS-004, LIS-005; SEC-006

- AC-01: Given an approved business and authorized active member, when valid food data, positive total quantity, unit, pickup location and valid deadline are submitted, then a donation is stored.
- AC-02: Given zero/negative quantity or an invalid date range, the API rejects the listing with structured validation errors.
- AC-03: An unapproved organization or unrelated user cannot publish the listing.

US-06 | Manage and cancel listing

Role: Restaurant / Bakery Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As business staff, I want to view, edit or cancel my listings so that availability remains accurate.

Requirement links: LIS-006, LIS-007; SEC-003, SEC-006; TRA-002

- AC-01: Authorized staff can retrieve and edit a listing owned by their organization when the requested transition is allowed.
- AC-02: When a donation is cancelled, the listing no longer accepts new NGO requests.
- AC-03: Changes to listing status record the actor and timestamp in audit history.

US-07 | Discover and filter donations

Role: Verified NGO Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As NGO staff, I want to browse and filter available donations so that I can identify suitable food.

Requirement links: LIS-008, LIS-009, LIS-010; USA-005

- AC-01: An authorized verified NGO can retrieve donation listings showing quantity, unit, location, category and deadline.
- AC-02: Filtering by supported location/category/quantity/deadline criteria returns only matching records; pagination follows MEAL-62.
- AC-03: Expired or cancelled listings cannot be requested even if a stale UI still displays them.

US-08 | Request a quantity

Role: Verified NGO Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As NGO staff, I want to request a portion of an existing donation so that my organization can collect a suitable amount.

Requirement links: REQ-001, REQ-002; REL-001, REL-003

- AC-01: Given an existing requestable donation and approved NGO, when requested_quantity is positive and within available quantity at submission, a pending request is created.
- AC-02: The request links the source donation, NGO organization and requesting user.
- AC-03: A request against an expired/cancelled donation, by an unverified NGO, or with nonpositive quantity is rejected.

US-09 | Review NGO requests

Role: Restaurant / Bakery Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As business staff, I want to see pending requests for my donations so that I can make allocation decisions.

Requirement links: REQ-003, REQ-006; SEC-003

- AC-01: An authorized member of the donation-owning business can see request quantity, NGO and status for its listings.
- AC-02: A business member cannot access private decision controls for another business's requests.
- AC-03: The UI accurately distinguishes pending, approved, rejected and cancelled request states.

US-10 | Approve full request safely

Role: Restaurant / Bakery Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As business staff, I want to approve the entire pending request so that food is reserved without oversubscription.

Requirement links: REQ-004, REQ-005, REQ-006; REL-002, REL-004; TRA-003

- AC-01: Given an authorized business user and a pending request, when sufficient allocatable quantity remains, approval marks that request approved and increments allocated_quantity by exactly requested_quantity in one transaction.
- AC-02: The API provides no partial-approval operation and does not accept an alternative approved quantity.
- AC-03: Given concurrent requests whose total exceeds remaining quantity, at most the quantity-fitting approval commits; the other receives a conflict and allocation never exceeds total_quantity.
- AC-04: The reviewer and decision timestamp are recorded, and failed transaction steps roll back all changes.

US-11 | Reject a request

Role: Restaurant / Bakery Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As business staff, I want to reject a pending NGO request so that the NGO receives an accurate decision.

Requirement links: REQ-004, REQ-006; TRA-003

- AC-01: Given a pending request for the business's donation, an authorized rejection sets status to rejected and records reviewer/time.
- AC-02: Rejecting does not increase or decrease allocated_quantity.
- AC-03: A second decision on a non-pending request is denied without altering allocation.

US-12 | Track status updates

Role: NGO / Business Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an involved user, I want visible updates so that I know what happened to my donation/request.

Requirement links: REQ-007; USA-002; REL-006; TRA-002, TRA-003

- AC-01: After request approval or rejection, the affected user sees the current state when next retrieving their records.
- AC-02: If in-app notifications are enabled, a notification is linked to the recipient and event and may be marked as read.
- AC-03: Only the intended recipient can access their personal notifications.

US-13 | View and accept assigned pickup

Role: Volunteer / Collector Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a collector, I want to view my assigned pickup so that I can arrive at the right place and time.

Requirement links: FUL-001, FUL-003; SEC-002; USA-004

- AC-01: An authorized collector can view pickup location, schedule, instructions and linked approved request for an assignment.
- AC-02: A pickup record can only be created or assigned for an approved request.
- AC-03: A collector cannot update a pickup assigned to a different collector unless authorized.

US-14 | Confirm collection and delivery

Role: Volunteer / Collector Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a collector, I want to update pickup/delivery status so that involved organizations can track progress.

Requirement links: FUL-001, FUL-002, FUL-004, FUL-007; TRA-004

- AC-01: When an allowed collection transition occurs, the system records picked_up status and collection timestamp.
- AC-02: When delivery is confirmed, the delivery status and timestamp are stored with valid chronological ordering.
- AC-03: Illegal transitions (e.g., delivered before picked_up) are rejected by the backend.

US-15 | Record final distribution

Role: Verified NGO Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As NGO staff, I want to record actual distribution so that impact is supported by evidence.

Requirement links: FUL-005, FUL-006, FUL-007; TRA-005; SEC-006

- AC-01: Given a successful associated pickup, authorized NGO staff can create one final distribution record linked to that pickup.
- AC-02: Distributed quantity is positive and no greater than the approved request's requested_quantity; beneficiary count, when provided, is nonnegative.
- AC-03: A second final distribution for the same pickup or distribution before successful pickup is rejected.

US-16 | Monitor and manage platform

Role: Administrator Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an administrator, I want to review organization, user, donation and request activity so that I can manage the platform.

Requirement links: ACC-008; FUL-008; SEC-002, SEC-004

- AC-01: An authenticated admin can view relevant management records and verification states.
- AC-02: A non-admin is denied admin-only API operations.
- AC-03: Displayed sensitive user data is limited to fields required for administrative functions.

US-17 | View confirmed impact reports

Role: Administrator Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an administrator, I want to view impact totals so that I can report meaningful results.

Requirement links: TRA-006; USA-006; REL-005

- AC-01: Reports calculate counts/quantities from persisted approved, collected or distributed records under explicitly named metric definitions.
- AC-02: Cancelled/rejected/unfulfilled records are excluded from metrics described as completed distributions.
- AC-03: Report figures are reproducible from the underlying database records.

US-18 | Enforce security and validation

Role: All roles Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a platform participant, I want data and actions protected so that records stay trustworthy.

Requirement links: SEC-002, SEC-003, SEC-004, SEC-005, SEC-006, SEC-007; REL-001, REL-004

- AC-01: Protected API actions validate the authenticated role and organization membership on the server, regardless of what the Angular UI hides.
- AC-02: Invalid quantities, deadlines and transitions return documented errors without partial database writes.

- AC-03: Application secrets are loaded from local environment configuration and are absent from tracked source files.

US-19 | Use accessible responsive screens

Role: All roles Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As a platform user, I want clear pages across typical browser sizes so that I can finish my task.

Requirement links: USA-001, USA-002, USA-003, USA-004

- AC-01: The role-specific view remains usable at common desktop and mobile-browser widths without horizontal clipping of primary controls.
- AC-02: Forms identify required fields and display understandable errors returned by the API.
- AC-03: Listing, request and pickup states are displayed as visible text labels rather than color alone.

US-20 | Maintain audit trail

Role: System / Administrator Priority: MUST for core MVP (except enhancements explicitly noted in source requirements)

As an administrator, I want traceable changes so that important decisions can be investigated.

Requirement links: TRA-001, TRA-002, TRA-003, TRA-004, TRA-005, TRA-007

- AC-01: Each story retains a stable unique story ID and mappings to requirement IDs.
- AC-02: Relevant approval, rejection, cancellation and pickup transitions record actor, old/new status and timestamp.
- AC-03: An audit event cannot silently claim a successful transition when its database transaction failed.

5. Traceability and test execution

Use story IDs US-01–US-20 in Jira tests or acceptance scenarios. Each AC is independently verifiable by UI/API or database observation. “Must” and “Should” remain authoritative at requirement-ID level in the Requirements Specification; this catalog does not silently elevate all optional suggestions into compulsory Sprint 2 delivery.

Area	Stories	Requirement families
Accounts / organizations	US-01–US-04	ACC, SEC
Donations / NGO requests	US-05–US-12	LIS, REQ, REL, TRA
Collection / distribution	US-13–US-15	FUL, TRA
Administration / reports	US-16–US-17	ACC, FUL, TRA
Cross-cutting	US-18–US-20	SEC, REL, USA, TRA

6. Development and test hand-off

- Build UI screens/components to satisfy user-facing acceptance conditions; do not rely on Angular route guards alone for access control.
- Implement MEAL-62 v1.3 API contracts in FastAPI; return documented HTTP status codes and error envelopes.
- Implement MEAL-6 migration constraints, organization authorization and transaction-safe quantity allocation in PostgreSQL/FastAPI.
- Write positive, negative, authorization and concurrency tests; link test case IDs to each US/AC and requirement ID.
- Story completion requires observed test evidence, not merely existence of a document or API skeleton.

7. Sources and design references

Meal Connect proposal and MVP Scope v1.1; Software Requirements Specification (61 requirement IDs across ACC/LIS/REQ/FUL/SEC/USA/REL/TRA); MEAL-5, MEAL-61, MEAL-6, MEAL-62, MEAL-63 v1.3; MEAL-66 ERD review. Some assignment/notification specifics are implementation conventions; use MEAL-62 as the single source for precise API requests/responses.